Juan Carlos Mendoza

Staff Security Engineer | Python Developer | AI & Automation

(509) 439-0227 | juanmendoza0309@gmail.com | linkedin.com/in/mendozajua | mendozajua.github.io/jmc-portfolio/

PROFESSIONAL SUMMARY

Staff Security Engineer and Python developer with 5+ years of experience building automation tools, AI-powered applications, and cloud infrastructure in enterprise security environments. Currently running an AWS Bedrock-powered web app for vulnerability scanning and leveraging MCP tooling to bring AI into daily security workflows. Leading the evaluation and adoption of an MCP-based SDE integration that will automate security survey updates across hundreds of app teams. Known for turning repetitive, manual processes into reliable automated systems and for building tools that teams actually use.

EXPERIENCE

Staff Security Engineer | **FINRA** *Rockville, MD • 2025 – Present*

- Built and maintain an AWS Bedrock-powered web application that scans SKILLS.md files for security vulnerabilities, delivering AI-driven analysis directly to development teams without requiring security expertise on their end.
- Identified, evaluated, and integrated existing MCP (Model Context Protocol) tooling into the team's security engineering workflows, meaningfully improving day-to-day productivity for the team.
- Built custom AI agents tailored specifically to security engineering job functions, leveraging MCP integrations to automate repetitive tasks and improve decision-making speed.
- Leading the adoption of an MCP integration from the SDElements vendor that will automatically scan application repositories and update security survey responses to reflect the true state of each codebase — reducing manual effort and improving accuracy of compliance data.
- Expanding the SDE MCP rollout to enable app teams to apply security countermeasures directly through the agent interface, closing the loop between vulnerability identification and remediation.
- Manage both the technical and business operations of the SDElements platform across 500+ applications, consistently achieving 90%+ quarterly survey completion through process automation and proactive team engagement.
- Refactored and stabilized the SDElements codebase, eliminating persistent recurring bugs and transforming a fragile system into a reliable, low-maintenance platform.
- Serve as the primary liaison between security requirements and engineering teams, translating compliance needs into practical developer guidance at all levels of the organization.

Cybersecurity Engineer | **FINRA** *Rockville, MD • Nov 2021 – 2025*

- Optimized AWS Lambda-based ETL pipelines for Splunk data ingestion, improving throughput by 50% and eliminating 1+ hour of weekly manual processing through automation.
- Built Jenkins CI/CD pipelines enabling one-click provisioning of AWS ECS-EC2, ECS-Fargate, and Lambda environments, reducing deployment setup time by 90%.
- Integrated Contrast Security agents across all Python applications via automated deployment workflows, cutting vulnerability remediation timelines from 1–6 months down to under 3 days.
- Designed a core Python application using OOP principles (classes, inheritance) to manage security metadata, reducing bugs by 25% and improving data integrity through optimized relational database queries.
- Built JSON-driven configuration systems and automated reporting scripts that reduced manual reporting overhead by 70% for a 1,400-person department.
- Developed automated email notification scripts that reduced team communication turnaround time by 92%.

Full Stack Developer | **Revature** *Reston, VA • Apr 2021 – Dec 2022*

- Led UI unit testing for a React/TypeScript application, achieving 80–90% code coverage and mentoring 19 bootcamp peers on testing best practices.
- Collaborated with a 19-person team to design and ship a quiz and study card application from concept to functional MVP within a two-week sprint.

Student Researcher | **Central Washington University** *Ellensburg, WA • Jun 2019 – Aug 2019*

- Built a Python-based influenza transmission simulation using differential equations, enabling faster parameter tuning and improving prediction accuracy by 15%.
- Produced 15+ Matplotlib visualizations supporting academic reporting and stakeholder presentations; proposed and championed the research direction that led to the project's completion.

Student Researcher | **Florida Institute of Technology** *Melbourne, FL • May 2018 – Jul 2018*

- Conducted ML research on academic publishing trends, analyzing over 1 million computer science papers using Python, Pandas, and NumPy.
- Built a data standardization pipeline that improved dataset consistency and ML model accessibility by approximately 30%.

PROJECTS

AI Agents & MCP Development *Personal • 2024 – Present*

- Built custom AI agents and MCP (Model Context Protocol) servers in personal time, exploring agentic workflows, tool-use patterns, and multi-step reasoning across a range of use cases.
- Hands-on experience designing agent architectures, defining MCP tool schemas, and integrating agents with external APIs and local data sources — skills directly applied to production tooling at work.

Home Lab — Secure Self-Hosted Infrastructure *Richland, WA • Jun 2023 – Present*

- Deployed Proxmox as a bare-metal virtualization platform, running 10+ VMs and Docker containers for self-hosted applications, isolated test environments, and automation workflows.
- Configured NGINX Proxy Manager, Cloudflare DNS across 10+ domains, and Authelia 2FA enforcement for layered access control across all exposed services.
- Set up VLANs for network segmentation and deployed 24/7 IP camera monitoring with secure local data storage.

EDUCATION

Bachelor of Science in Applied Mathematics *Sept 2015 – Mar 2020*

Central Washington University, Ellensburg, WA

TECHNICAL SKILLS

Languages: Python, Bash/Shell Scripting, Java, TypeScript, JavaScript, R, Groovy, HTML, CSS

AI & Agents: AWS Bedrock, MCP (Model Context Protocol), AI Agent Development, LLM Integration, Prompt Engineering

Cloud & DevOps: AWS (Lambda, ECS, EC2, Fargate, S3, Bedrock), Jenkins, Docker, Git, CI/CD Pipelines

Security Tools: SDElements, Contrast Security, Splunk, Authelia, Vulnerability Management, Secure Code Practices

Data & Automation: ETL Pipelines, Pandas, NumPy, JSON config systems, REST API Integration, Postman, DBeaver

Infrastructure: Proxmox, NGINX Proxy Manager, Cloudflare, VLANs, Kubernetes (foundational)

Frameworks: React, Spring Boot, Swagger UI, Matplotlib

CERTIFICATIONS & AWARDS

- AWS Solutions Architect Associate – Certification
- AWS Cloud Practitioner – Certification
- Solver Scholarship – $20,000 funded by NSF S-STEM Foundation
- Hearst-SOLVER Fellowship – $4,000 research grant